

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Summer Examination – 2024

Course: B. Tech. Branch: Common to All branches Semester: II

Subject Code & Name: BTES205E - Energy and Environment Engineering

Max Marks: 60

Date: 24/06/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Short Notes:- i) Conventional Energy Sources ii) Steam Power Generation	CO1	6
B) Explain in detail Gas Turbine Power Plant with neat Diagram.	CO2	6
C) Difference between :- Thermo electric and Thermionic generators.	CO3	6
Q.2 Solve Any Two of the following.		12
A) Short Notes:- i) Biogas ii) Biomass	CO3	6
B) Explain the Fuel Cell with neat diagram.	CO2	6
C) Explain in detail Magneto Hydro Dynamics (MHD) with neat diagram.	CO1	6
Q. 3 Solve Any Two of the following.		12
A) Short Notes:- i) Maximum energy efficiency ii) Maximum cost effectiveness.	CO1	6
B) Write the Methods and techniques of energy conservation in ventilation and air conditioners, compressors, pumps, fans and blowers.	CO2	6
C) Write the Energy conservation in electric furnaces, ovens and boilers. lighting techniques.	CO3	6
Q.4 Solve Any Two of the following.		12
A) Write in detail about Air pollution Sources and Effects.	CO3	6
B) Write in detail the control measures in Air pollution.	CO4	6
C) Short Notes:- i) Particulate Emission ii) Air quality standards	CO5	6

Q. 5 Solve Any Two of the following.

12

A) Write the Water Pollution Effects and Control Measures.

CO3

6

B) Write the Noise pollution effects and control measures.

CO4

6

C) Short Notes :- i) Thermal Pollution ii) Soil Pollution

CO5

6

***** End *****